

An Agentic Orchestration Layer for Machine-Learning Gravitational-Wave Analysis: Typed Skill Contracts, Fail-Closed Execution and Calibrated Multi-Model Inference

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ABSTRACT

Machine-learning pipelines now cover the full low-latency gravitational-wave workflow: detection (Aframe), parameter estimation (AMPLFI), unmodelled anomaly search (GWAK) and noise subtraction (DeepClean). Composing them for a given scientific request still requires expert judgement about data access, model applicability, significance calibration and compute. We present ML4GW-AGENT, an orchestration layer in which a large language model or a deterministic planner turns a natural-language request into a directed acyclic graph of typed skills, each bound to a versioned contract (JSON-schema inputs and outputs, machine-checked preconditions, immutable model revisions) and executed by a fail-closed adapter that records every artifact by cryptographic hash. We evaluate the layer on three axes: planning quality on a benchmark of NNN prompts across three model backbones, including an adversarial suite that measures silent, plausible-but-wrong outcomes; scientific fidelity on the 90 events of GWTC-1, GWTC-2.1 and GWTC-3 processed end to end through Aframe, AMPLFI and GWAK on the LIGO Data Grid, compared with the catalog and with the published Aframe O3 search; and significance calibration from time-shifted backgrounds of NNN days (Aframe) and NNN days (GWAK) together with injection campaigns. TBD results. The code, contracts, calibration tables and run manifests are public.

Keywords: Gravitational waves — Neural networks — Astronomy software — Astronomy data analysis

1. INTRODUCTION

TBD: ML4GW pipelines in production (Aframe live since 2025-08; AMPLFI; GWAK; DeepClean). Agentic AI for science: Coscientist, ChemCrow, Virtual Lab, Co-Scientist; in astronomy Mephisto (ApJS 2026), CM-BAgent/Denario. The reliability problem: silent incorrect computation (?) and fabricated results in gw-Benchmarks (?). Our thesis: contracts and fail-closed adapters make an agent’s science auditable.

2. DESIGN

2.1. Skill contracts

2.2. Planning: deterministic baseline and LLM planner

2.3. Fail-closed adapters and provenance

2.4. Calibration as a first-class object

2.5. Executors and cost policy

3. EVALUATION OF PLANNING

3.1. Benchmark

3.2. Backbones

3.3. Silent-failure suite

4. SCIENTIFIC VALIDATION ON GWTC EVENTS

4.1. Setup on the LIGO Data Grid

4.2. Detection: comparison with the catalog and with the Aframe O3 search

4.3. Parameter estimation: AMPLFI versus GWTC posteriors

4.4. Anomaly scores

5. SIGNIFICANCE CALIBRATION

5.1. Time-shifted backgrounds

5.2. Injections

5.3. DeepClean applicability and signal preservation

6. COST AND SCALABILITY

7. DISCUSSION

8. CONCLUSIONS

1 TBD. This research has made use of data or software ob-
2 tained from the Gravitational Wave Open Science Cen-
3 ter (gwosc.org), a service of the LIGO Scientific Collab-
4 oration, the Virgo Collaboration, and KAGRA.

Software: ml4gw-agent, Buoy, Aframe, AMPLFI,
GWAK, DeepClean, ml4gw, gwpy, astropy, PyTorch,
HTCondor